

Problem 1

Problem 1

**Problem 1** Fill in the following blanks with the correct choice of the words from this list:

Increasing, decreasing, positive, negative, concave up, concave down

- (a) If you know  $f''(x) > 0$ , then you know  $f'(x)$  is \_\_\_\_\_  
and  $f(x)$  is \_\_\_\_\_.
- (b) If you know  $g'(x) < 0$  and decreasing, then you know  $g(x)$  is \_\_\_\_\_  
and \_\_\_\_\_.
- (c) If you know  $h(x)$  is positive, increasing, and concave down, then you know  
 $h'(x)$  is \_\_\_\_\_ and \_\_\_\_\_  
and that  $h''(x)$  is \_\_\_\_\_.

[4]